

cases that have been talked about for nearly a decade now.

Supply chain management:

Blockchain can be used to track the movement of goods and materials through a supply chain. This can help to improve efficiency, transparency, and traceability.

Healthcare: Blockchain can be used to store and share medical records in a secure and private way. This can help to improve patient care, and to reduce fraud and errors.

Government: Blockchain can be used to create secure and transparent records of government transactions. This can help to improve accountability and transparency, and to reduce corruption.

Other industries: Blockchain is also being explored for use in other industries, such as insurance, real estate, and media.

The current state of blockchain technology outside of crypto and finance

Outside of cryptocurrencies, most of which are directly or indirectly under the investigation of SEC, not much has happened for blockchain technology. The total locked value (TVL) of decentralised financing went from an all-time high of \$248 billion in 2022 to less than \$40 billion in 2023, and NFTs cooled off almost as fast as they had heated.

I did a survey of large scale applications of blockchain technologies in 2023 and I am afraid I could not find much to talk about. Keep in mind that this is my own personal research and therefore it's not conclusive. I might have missed an application or two, but overall, large scale applications (not related to crypto and finance) currently running on blockchain technology are next to non-existent.

Current applications of blockchain across the world

IBM Food Trust: IBM Food Trust is a blockchain-based platform

that tracks the movement of food products through the supply chain. Launched in 2021, this platform is used by food producers, retailers, and distributors to ensure the safety and authenticity of food products.

Walmart: Walmart is experimentally using blockchain technology to track the movement of produce through its supply chain at a limited capacity. Not much has been spoken about this project since its inception in 2022.

Estonia: Estonia is using blockchain technology to record land ownership and manage elections. It is one of the most advanced countries in the world in terms of its use of blockchain technology.

United Arab Emirates: The United Arab Emirates is using blockchain technology experimentally to track the movement of goods and materials through its ports.

The 'big problem' for blockchain

For most blockchain experts, the big problem to solve is the 'blockchain triad', as coined by Vitalik Buterin, wherein every blockchain application has to choose between focusing on decentralisation, security or scalability but cannot have all three without compromising on at least one.

But I would like to argue for a bigger 'big problem' for blockchain — adaptability. Outside of Bitcoin, which seems to be the only technology to find its grounding in a proper use case and an application, blockchain is struggling. We are yet to put theory into practice when it comes to blockchain and this is becoming clearly evident with every passing day.

Probable reasons for the blockchain adoption abyss

As a researcher, I can only share potential reasons for the lack of adoption of blockchain technology based on observational data. This is by no means an exhaustive list of reasons, and I

welcome any feedback and criticism on the points that I have raised here.

Complexity and understanding:

Not only does blockchain suffer from bad PR due to crypto and NFTs, it also has a problem with outreach. Most people, including the ones invested in blockchain technology, do not understand how the technology works and what makes it uniquely useful. A survey by *Statista* in 2022 found that only 39% of Americans have heard of blockchain. Of those who have heard of it, only 16% said they have a good understanding of it.

Lack of investment by big tech:

Although tech startups disrupt industries more often than big tech, it is often the big tech companies that throw money behind such companies and projects to fuel the disruption. A classic example is Microsoft with OpenAI. Unfortunately, such a thing has not happened in the case of blockchain. Despite there being no shortage in blockchain startups, big tech companies such as Google, Microsoft and Meta have not been very forthcoming in investing in these projects. In fact, the world of big tech has been suspiciously silent on the whole blockchain saga. There are multiple reasons for this.

- **The crypto contagion:** The most valuable asset of a big tech company is its stock price. Protecting the value of its stock is the first priority for the CEOs of such companies. Therefore, getting into a technology that has direct ties with the crypto space can be potentially damaging for the stock price.
- **Lack of regulatory clarity:** Even if an interesting project worth investing were to come up, it is difficult for cryptocurrencies to not play some form of a role in that project. Given the lack of regulatory clarity from the SEC, as it stands today, the future of blockchain projects can be highly speculative. Big tech likes due diligence when